ALAMEIN INTERNATIONAL UNIVERSITY
جامعة العلمين الدولية

Phy 211

Lecture 5 Notes

By

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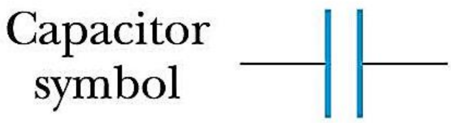
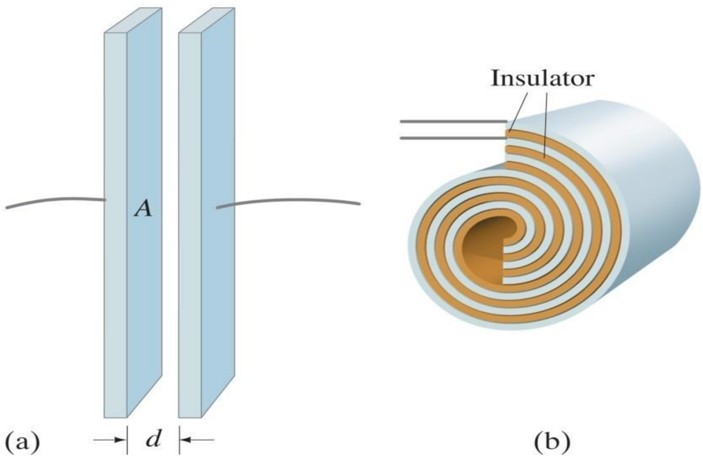
Chapter 4
Capacitors

➢ Capacitors "Condensers":

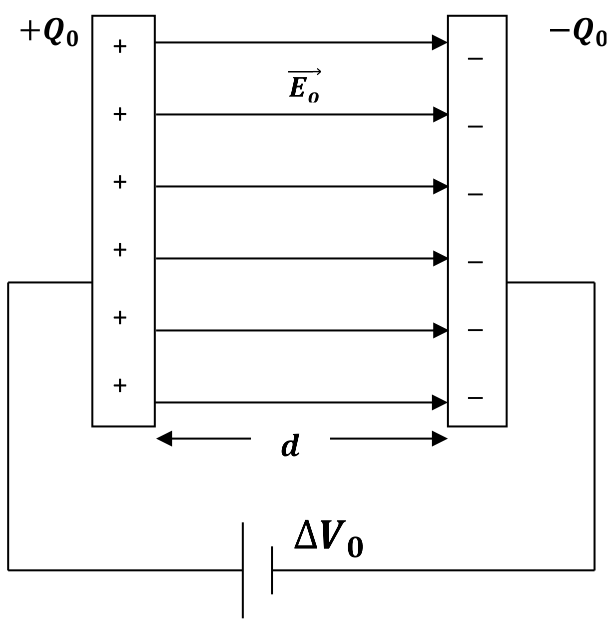
- **Capacitor:** It is a device that can store electric charge. It consists of two metal conductors placed near each other but not touching.

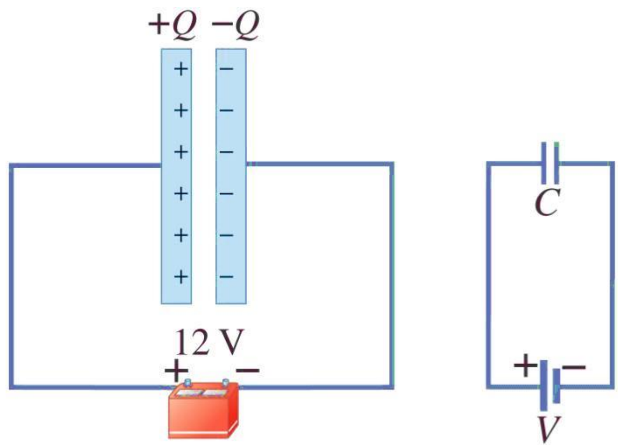




a) Parallel plate capacitor b) Another capacitor type

- **Charging a parallel plate capacitor:**





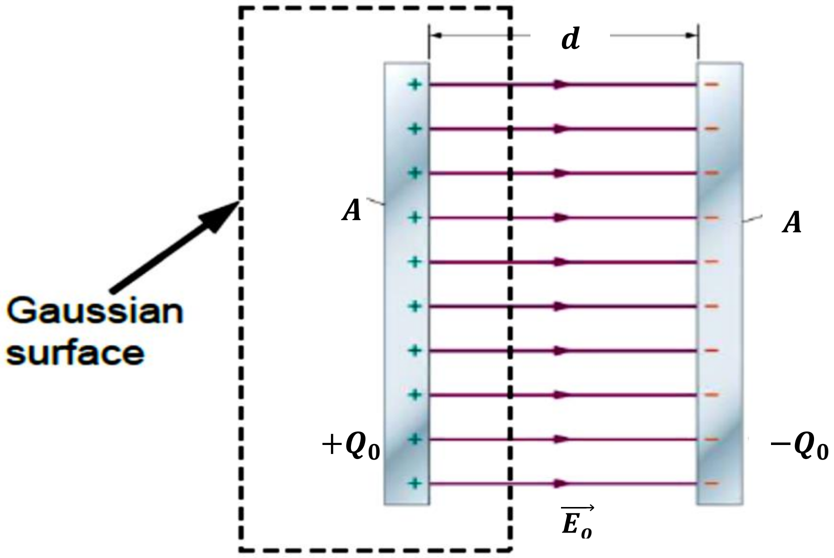
$$\boxed{E_0 = \Delta V_0/d}$$

- If a potential difference "ΔV_0" is applied to a parallel plate capacitor, it quickly becomes charged, the amount of charge "Q_0" on each plate is directly proportional to ΔV_0, therefore

$$Q_0 \propto \Delta V_0$$

$$\boxed{\therefore \boldsymbol{Q_0 = C_0 \Delta V_0} \longrightarrow \boldsymbol{C_0 = \frac{Q_0}{\Delta V_0}}}$$

- **Capacitance "C":** It is the ratio of the magnitude of the charge on either conductor to the magnitude of the potential difference between the conductors. $\left(Unit: \frac{\boldsymbol{Coulomb}}{\boldsymbol{Volt}} = \mathbf{F}(\boldsymbol{farad})\right)$



From Gauss' law: $$\emptyset_E = E_0 A = \frac{Q_0}{\varepsilon_o}$$

$$\frac{\Delta V_0}{d} A = \frac{C_0 \Delta V_0}{\varepsilon_o}$$

$$\boxed{\therefore \boldsymbol{C_0 = \frac{\varepsilon_o A}{d}}}$$

$$\boldsymbol{C \propto A(area\ of\ each\ plate)} \quad \& \quad C \propto \frac{\mathbf{1}}{\boldsymbol{d(distance\ between\ 2\ plates)}}$$

- **Example 1:** a parallel plate capacitor has an area A and separation d, what will be happened if

1. The area is doubled
2. The separation is halved
3. The area is doubled and the separation is halved

Solution

$$\mathbf{1.}\ \boldsymbol{C_1 = \frac{\varepsilon_o A}{d}} \quad \& \quad \boldsymbol{C_2 = \frac{\varepsilon_o 2A}{d}} \longrightarrow \boldsymbol{\frac{C_2}{C_1} = \frac{2\,\frac{\varepsilon_o A}{d}}{\frac{\varepsilon_o A}{d}} = 2} \longrightarrow \boldsymbol{C_2 = 2C_1}$$

$$\mathbf{2.}\ \boldsymbol{C_1 = \frac{\varepsilon_o A}{d}} \quad \& \quad \boldsymbol{C_2 = \frac{\varepsilon_o A}{\frac{1}{2}d}} \longrightarrow \boldsymbol{\frac{C_2}{C_1} = \frac{2\,\frac{\varepsilon_o A}{d}}{\frac{\varepsilon_o A}{d}} = 2} \longrightarrow \boldsymbol{C_2 = 2C_1}$$

$$\mathbf{3.}\ \boldsymbol{C_1 = \frac{\varepsilon_o A}{d}} \quad \& \quad \boldsymbol{C_2 = \frac{\varepsilon_o 2A}{\frac{1}{2}d}} \longrightarrow \boldsymbol{\frac{C_2}{C_1} = \frac{4\,\frac{\varepsilon_o A}{d}}{\frac{\varepsilon_o A}{d}} = 4} \longrightarrow \boldsymbol{C_2 = 4C_1}$$

- **Example 2:** Two conductors having net charges of $+10.0\ \mu\text{C}$ and $-10.0\ \mu\text{C}$ have a potential difference of $10.0\ \text{V}$ between them.

(a) Determine the capacitance of the system.

(b) What is the potential difference between the two conductors if the charges on each are increased to $+100\ \mu\text{C}$ and $-100\ \mu\text{C}$?

Solution

$$Q = 10\ \mu\text{C} \qquad \Delta V = 10\ \text{V}$$

$$a)\quad C = \frac{Q}{\Delta V} = \frac{10 \times 10^{-6}}{10} = 10^{-6}\ \text{F} = 1\ \mu\text{F}$$

$$b)\quad \Delta V = \frac{Q}{C} = \frac{100 \times 10^{-6}}{1 \times 10^{-6}} = 100\ \text{V}$$

- **Example 3:**

a) Calculate the capacitance of a parallel-plate capacitor whose plates are 20 cm by 3 cm and are separated by a 1 mm air gap.

b) What is the charge on each plate if the capacitor is connected to a 12 V battery?

c) What is the electric field between the plates?

Solution

$$a)\ \ C = \frac{\varepsilon_o A}{d} = \frac{(8.85 \times 10^{-12}) \times (0.2 \times 0.03)}{1 \times 10^{-3}} = 53 \times 10^{-12}\ \mathrm{F} = 53\ \mathrm{pF}$$

$$b)\ \ Q = C\Delta V = (53 \times 10^{-12}) \times (12) = 636 \times 10^{-12}\ \mathrm{C}$$

$$c)\ E = \frac{\Delta V}{d} = \frac{12}{1 \times 10^{-3}} = 12000 \left(\frac{\mathrm{V}}{\mathrm{m}}\ or\ \frac{\mathrm{N}}{\mathrm{C}}\right)$$

➢ Energy stored in a charged capacitor

- The work done in charging the capacitor (to move +ve & -ve charges to the plates) appears as electric potential energy (U) stored in the capacitor.

$$\boxed{\boldsymbol{U = \frac{Q^2}{2C} = \frac{1}{2} Q\Delta V = \frac{1}{2} C(\Delta V)^2}}\ (\mathbf{Joule})$$

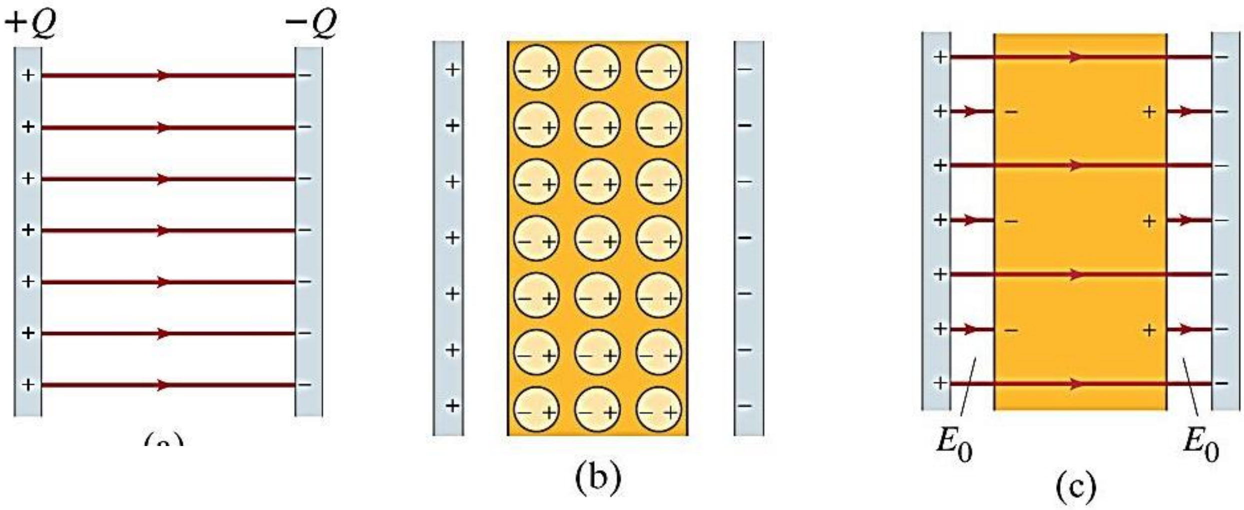
- **Storage Energy density (u):** It is the energy stored per unit volume

$$\boldsymbol{u = \frac{U}{Volume} = \frac{U}{Ad} = \frac{\frac{1}{2} C(\Delta V)^2}{Ad} = \frac{1}{2}\frac{\left(\frac{\varepsilon A}{d}\right)(Ed)^2}{Ad}}$$

$$\boxed{\boldsymbol{u = \frac{1}{2}\varepsilon E^2 \quad \left(\frac{\mathbf{Joule}}{\mathbf{m^3}}\right)}}$$

➢ Dielectric:

A dielectric is an insulating material that, when placed between the plates of a capacitor, increases the capacitance.



(a) (b) (c)

If the capacitor plates is separated by air then

$$C_o = \frac{Q_0}{V_o} \quad \& \quad E_o = \frac{\Delta V_o}{d}$$

If a dielectric is placed in between the plates, the molecules of the dielectric will be polarized inducing an electric field in opposite direction to E_o so that the electric field will be reduced by factor k.

$$\boxed{\boldsymbol{E = E_o - E' = \frac{E_o}{k}}} \longrightarrow \boxed{\boldsymbol{\therefore \Delta V = \frac{\Delta V_o}{k}}} \ \& \ \boxed{\boldsymbol{Q = Q_0}}$$

$$\therefore C = \frac{Q}{V} = \frac{Q_0}{\Delta V_o / k} = k\frac{Q_0}{\Delta V_o}$$

$$\boxed{\boldsymbol{\therefore C = kC_o}} \ \boldsymbol{capacitance\ increases}$$

Therefore, the capacitance increased by a factor k (dielectric constant of material "unitless")($k \geq 1$)

$$\boxed{\therefore \boldsymbol{C = \frac{k\varepsilon_o A}{d} = \frac{\varepsilon A}{d}}}$$

ε: electric permittivity of material.

$$\left(\boldsymbol{\varepsilon = k\varepsilon_o \longrightarrow k = \frac{\varepsilon}{\varepsilon_o} = \varepsilon_r\ relative\ permittiviy}\right)\ (k = 1\ for\ air)$$

- **Example 4:**

A parallel plate capacitor is constructed with plates of area $0.028\ \mathrm{m}^2$ and separation $0.55\ \mathrm{mm}$. The space between the plates is filled with a dielectric of dielectric constant k. When the capacitor is connected to a $12\ \mathrm{V}$ battery, each of the plates has a charge of magnitude $3.62 \times 10^{-8}\ \mathrm{C}$. What is the value of the dielectric constant k?

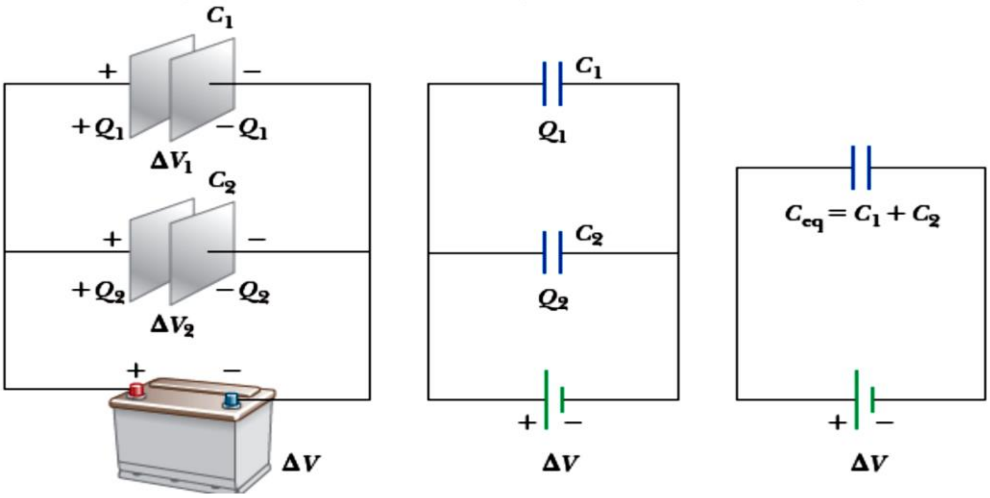
Solution

$$\because C = \frac{k\varepsilon_o A}{d} \longrightarrow \frac{Q}{\Delta V} = k\ \frac{\varepsilon_o A}{d}$$

$$k = \frac{Qd}{\Delta V \varepsilon_o A} = \frac{(3.62 \times 10^{-8})(0.55 \times 10^{-3})}{(12)(8.85 \times 10^{-12})(0.028)} = 6.7$$

➢ Combinations of Capacitors:

- **Capacitors in parallel:**



$$\boxed{\Delta V = \Delta V_1 = \Delta V_2}$$

$$\boxed{Q_{total} = Q_1 + Q_2}$$

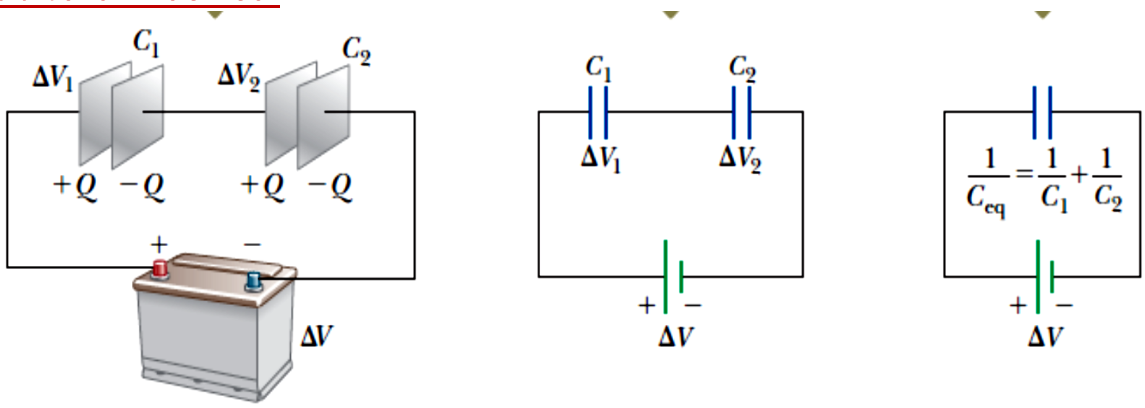
$$\boxed{C_{eq} = C_1 + C_2} \quad \textit{total capacitance increases}$$

$$\boxed{Q_{total} = C_{eq}\Delta V}$$

For n identical capacitors each of capacitance C connected in parallel:

$$C_{eq} = nC$$

- **Capacitors in series:**



$$\boxed{\Delta V = \Delta V_1 + \Delta V_2 + \cdots}$$

$$\boxed{Q_{total} = Q_1 = Q_2 = \cdots = Q}$$

$$\boxed{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots} \quad \textit{total capacitance decreases}$$

$$\boxed{Q_{total} = C_{eq}\Delta V}$$

For n identical capacitors each of capacitance C connected in series:

$$C_{eq} = C/n$$